

**Synthetic Expression Analysis of Protein X Under Stress
Conditions**

Abstract

We report the expression of Protein X across multiple stress conditions. Western blot and microscopy analyses (Figure 1, Figure 2) demonstrate condition-dependent modulation. Quantification indicates a 2.5-fold increase relative to control ($p < 0.001$, $n = 6$ independent replicates).

Methods

Cells were cultured under standard conditions. Protein was extracted and separated by SDS-PAGE. Western blots were performed with $n = 6$ independent biological replicates per condition. Band intensities were quantified with ImageJ. Error bars in all figures represent mean $\pm$ SEM. Statistical significance was assessed by two-tailed t-test.

Results

Protein X expression increased markedly under stress. As shown in Figure 1, we resolved 12 independent lanes corresponding to 12 treatment conditions, each showing distinct band intensity. Densitometry revealed a 2.5-fold increase over control ($p < 0.001$). Figure 2 shows representative microscopy fields with consistent sub-cellular localization across all $n = 6$ replicates.

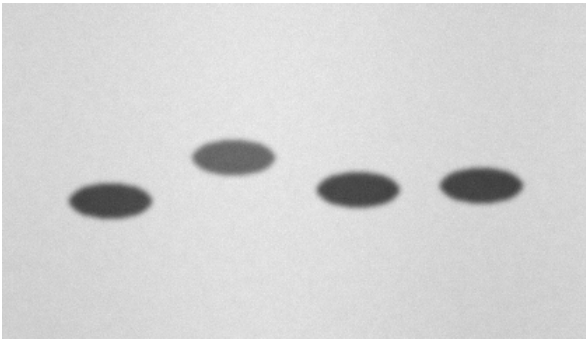


Figure 1. Western blot of Protein X across 12 treatment conditions ($n = 12$ lanes). Band intensity increases with stress severity. Error bars represent mean $\pm$ SEM.

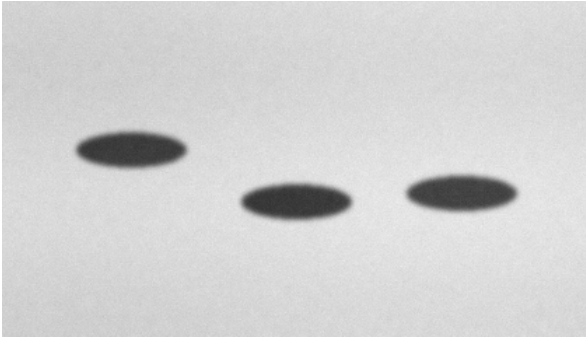


Figure 2. Representative microscopy fields showing sub-cellular localization of Protein X (n = 6 replicates, 3 conditions shown).

Discussion

These findings suggest Protein X is stress-responsive. The magnitude of the effect (2.5-fold, $p < 0.001$) is consistent with prior reports. Limitations include the synthetic nature of this test dataset.